
Enhancing Human-Machine Collaboration in Industrial Workflows using Generative AI Models

DISSERTATION

Submitted in partial fulfillment of the requirements for the Degree:
M.Tech. in Artificial Intelligence & Machine Learning

By

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AIMLCZG628T DISSERTATION

Dissertation Title: Enhancing Human-Machine Collaboration in Industrial Workflows using Generative AI Models

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Courses Relevant for the Project & Corresponding Semester:

Sl. No.	Course Name	Course ID	Semester	Year
1	Machine Learning	AIMLCZG565	1 st	2023–24
2	Deep Neural Networks	AIMLCZG511	2 nd	2023–24
3	Fair Accountable Transparent Machine Learning	AIMLCZG517	2 nd	2023–24
4	Advanced Deep Learning	AIMLCZG513	1 st	2024–25

Abstract

Modern industrial environments, including manufacturing, oil and gas, and energy industries, are becoming increasingly complicated and automated, requiring extremely effective human-machine collaboration. Although existing Artificial Intelligence (AI) models have considerably improved human capabilities, attaining seamless synergy is hampered by intrinsic industrial data restrictions, specifically a lack of anomaly or failure data. Traditional Machine Learning (ML) and non-generative Deep Learning (DL) models, which are inherently discriminative, struggle to address these real-world data limitations because they cannot predict such circumstances. To address these limitations, this study analyzes the transformational potential of Generative Artificial Intelligence (GenAI) techniques other than Transformer based models.

Existing industrial processes often rely on predictive models trained on limited historical anomaly data, leading to suboptimal performance in rare event detection and a lack of com-

prehensive scenarios for operator training. To address this, the work first explores the critical role of Variational Autoencoders (VAEs) in robust anomaly detection within industrial data. Industrial systems frequently operate within a narrow range of normal parameters, making anomaly instances exceedingly rare. VAEs, as a generative model, are uniquely suited to learn the complex underlying distribution of normal operational states. VAEs can effectively identify deviations as anomalies, even when no or minimal actual anomaly data are available for training. This capability is pivotal for developing reliable early warning systems that improve the situational awareness of human operators and enable proactive decision-making in safety-critical contexts.

However, while abundant in normal operational states, real-world industrial data often lack the diverse and representative samples needed for comprehensive training and rigorous validation of advanced AI models and human operators. To address it, this work shifts its focus to synthetic data generation for training and validation, applying other GenAI methodologies, viz. Generative Adversarial Networks (GANs). The work leverages the framework for its specific strengths. GANs, through adversarial training, excel at producing highly realistic and diverse synthetic samples that mimic complex time-series data or rare failure scenarios.

VAEs and GANs have the inherent potential to generate unique, coherent, and diverse data, in contrast to standard ML and non-generative DL techniques, which are discriminative and limited to learning from existing data. This generative capability is critical for overcoming the constraints of scarce or unbalanced industrial datasets, simulating unobserved scenarios, and providing an unparalleled means of robustly training AI systems and human personnel in difficult, data-poor industrial environments. This dissertation intends to demonstrate how these methods for generating and interpreting industrial data directly contribute to more confident decision-making in crucial scenarios by boosting human situational awareness. Consequently, enhancing human collaboration with reliable AI systems in complex industrial landscapes, thereby driving efficiency and safety in the era of Industry 4.0.

Key Words: Data Scarcity, Industrial Data, Anomaly Detection, Synthetic Data Generation, Generative AI

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Dissertation Outline

BITS ID No.: 2023AA05218

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Topic of Dissertation: Enhancing Human-Machine Collaboration in Industrial Workflows using Generative AI Models



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1. Project Work Overview

Title: Enhancing Human-Machine Collaboration in Industrial Workflows using Generative AI Models

1.1. Purpose

This work aims to enhance human-machine collaboration in industrial settings by addressing critical data limitations through GenAI. Specifically, it develops methods for robust anomaly detection using VAEs and generates high-fidelity synthetic industrial data using GANs. The ultimate purpose is to improve AI system training, validate their reliability, and create realistic environments for the preparation of human operators, promoting safer and more efficient operations.

1.2. Expected Outcome

- ⇒ Development of highly effective anomaly detection systems for industrial data, capable of identifying critical events even with minimal real-world anomaly data.
- ⇒ Creation of realistic and diverse synthetic industrial datasets for various operational scenarios, including rare failures and complex events.
- ⇒ Improved generalization and robustness of downstream machine learning models (e.g., for predictive maintenance)
- ⇒ A framework for rigorous validation of AI-driven control and assistance systems under a wide array of simulated conditions, boosting confidence in their real-world deployment.

1.3. Literature Review

1. Radicioni, L.; Bono, F.M.; Cinquemani, S. *“Vibration-Based Anomaly Detection in Industrial Machines: A Comparison of Autoencoders and Latent Space”*. *Machines* **2025**, *13*, 139.

This paper compares autoencoder-based methods for vibration-based anomaly detection in industrial machines, highlighting the limitations of reconstruction-based metrics for Variational Autoencoders (VAEs) and suggesting the need for latent space metrics.

2. A. A. Pol, V. Berger, C. Germain, G. Cerminara and M. Pierini, *“Anomaly Detection with Conditional Variational Autoencoders”*, 2019 18th IEEE International Conference On Machine Learning And Applications (ICMLA), Boca Raton, FL, USA, 2019, pp. 1651-1657.

This research introduces a method for conditional anomaly detection in production environments using Variational Autoencoders (VAEs). It focuses on providing interpretable *interesting scores* under real-time industrial constraints.

3. Keskes, Mohamed. (2025). “*Generative Adversarial Networks for Synthetic Data Generation in Deep Learning Applications*”, Preprints, 2025. 10.20944/preprints202505.0836.v1.

This review offers a comprehensive analysis of Generative Adversarial Networks (GANs) for synthetic data generation in deep learning applications. It explores their principles, advanced architectures like TimeGANs for time-series data, and their role in addressing data scarcity and privacy.

4. H. Lu, M. Du, K. Qian, X. He and K. Wang, “*GAN-Based Data Augmentation Strategy for Sensor Anomaly Detection in Industrial Robots*” in *IEEE Sensors Journal*, vol. 22, no. 18, pp. 17464-17474, 15 Sept.15, 2022.

This paper proposes a GAN-based data augmentation strategy to improve sensor anomaly detection in industrial robots. It addresses the challenge of scarce failure data by generating synthetic samples for robust model training.

1.4. Existing Process & Limitations

1. Anomaly Detection

(a) Statistical Methods

- ⇒ **Methodology:** Identifies data points deviating significantly from statistical norms (e.g., beyond a certain standard deviation or inter-quartile range).
- **Limitations:** They often rely on assumptions about data distributions (e.g., Gaussian), which rarely hold true for complex industrial data. And they also struggle with evolving data patterns (concept drift) and complex, non-linear relationships.

(b) Machine Learning Techniques

i. Clustering

- ⇒ **Methodology:** Groups similar data points into clusters and points far from any cluster centroid or in sparse regions are treated as anomalous.
- **Limitations:** They are sensitive to hyper-parameter tuning (e.g. number of clusters for K-Means) and they do not provide a probabilistic measure of abnormality or uncertainty.

ii. Decision Forests

- ⇒ **Methodology:** Decision forests like Isolation Forest builds an ensemble of isolation trees that recursively partition data, and anomalies, being *isolated*, require fewer partitions to be separated.
- **Limitations:** They are also sensitive to hyper-parameter tuning (e.g. contamination factor).

iii. Classification

- ⇒ **Methodology:** Algorithms like One-Class SVM learns a decision boundary around the *normal* data points in a high-dimensional space, and any point outside this boundary are classified as anomalous.
- **Limitations:** One-Class SVM suffer from the *curse of dimensionality*, therefore, they are computationally expensive for large datasets.

(c) Non-generative Deep Learning Techniques

i. Autoencoders (AEs):

- ⇒ **Methodology:** Trained to reconstruct *normal* input data and anomalies result in high reconstruction errors. (While VAEs are a type of AE, standard AEs are non-generative in the sense they do not learn a probabilistic latent space of the input data points).
- **Limitations:** Often require large amounts of labeled *normal* data for effective training, and their performance can degrade if normal data are not diverse enough.

ii. Recurrent Neural Networks (RNNs)

- ⇒ **Methodology:** Trained to predict the next value in a sequence and large prediction errors indicate an anomaly.
- **Limitations:** Can easily overfit to the training data, leading to poor generalization on unseen anomalies.

2. Synthetic Data Generation

(a) Statistical Methods

- ⇒ **Methodology:** Generating data based on statistical distributions (e.g., Gaussian, Uniform) derived from real data properties.
- **Limitations:** Cannot capture the complex, multivariate, and non-linear relationships present in real-world industrial data.

(b) **Rule-Based Methods**

- ⇒ **Methodology:** Defining explicit rules or patterns to generate synthetic data points.
- **Limitations:** Rule-based systems require extensive domain expertise and manual effort to define rules, which can be prone to human bias and incompleteness.

1.5. Justification for Methodology

- ✓ **Addressing Data Scarcity:** Generate realistic data for rare anomalies and failure scenarios, overcoming the critical limitation of insufficient real-world data for robust training.
- ✓ **Modeling Complex Distributions:** VAEs and GANs can explicitly learn and replicate the intricate, non-linear patterns inherent in industrial data, beyond what discriminative models can capture.
- ✓ **Enabling Comprehensive Training:** Create diverse synthetic environments for AI models and human operators, allowing exposure to a vast range of scenarios (including hazardous ones) that are impossible or too costly to replicate physically.
- ✓ **Enhancing System Validation:** Provide a rich, controllable dataset for rigorously validating AI system performance under varied conditions, leading to more robust and trustworthy deployments.

1.6. Project Work Methodology

- ⇒ **Phase 1: Project Setup & Data Wrangling:** Establish the project foundation, conduct initial research, collect and prepare data for modeling.
- ⇒ **Phase 2: VAE Model Development for Anomaly Detection:** Implement, optimize, and evaluate VAEs for robust anomaly detection.
- ⇒ **Phase 3: GAN Model Development & Synthetic Data Generation:** Implement, optimize, and evaluate GANs for generating high-fidelity synthetic industrial data.
- ⇒ **Phase 4: Dataset Augmentation & ML Model Generalization:** Use generated synthetic data to enhance the performance and generalization of existing machine learning models.

1.7. Benefits Derivable from the Work

- ✓ **Robust Anomaly Detection:** The use of generative models effectively overcomes the limitation of minimal or no available anomaly data of industrial processes. This allows for accurate detection of deviations, a task traditional methods struggle with when anomaly labels are rare.
- ✓ **Synthetic Industrial Data Generation:** Generative models allow for the synthesis of data representing rare failure modes, edge cases, and complex operational scenarios that are difficult, unsafe, or costly to capture or replicate in reality. This significantly expands the scope of data available for analysis and training.
- ✓ **Overcoming Data Privacy/Security Constraints:** By generating synthetic data that retains statistical properties but not sensitive original information, these methodologies offer a solution for developing and testing AI models in contexts where real industrial data cannot be directly shared or used due to privacy or security concerns.
- ✓ **Improved Human Situational Awareness:** By providing AI models trained on diverse synthetic data, the system can better highlight critical events, allowing human operators to quickly identify and resolve operational issues, minimizing downtime and optimizing processes.
- ✓ **Increased Human Trust in AI Systems:** The ability to thoroughly validate AI models using diverse synthetic data scenarios before deployment leads to more reliable AI. This demonstrable reliability fosters greater trust from human operators, encouraging effective collaboration.
- ✓ **Reduced Cognitive Load for Operators:** By enabling AI to proactively identify subtle anomalies or predict potential issues, the system can reduce the cognitive burden on human operators, allowing them to focus on higher-level decision-making and problem-solving.

2. Broad Area of Work

The work intersects advanced Artificial Intelligence (AI) and critical industrial applications. It directly addresses the critical pain point of data scarcity, particularly for rare anomalies and failure events, in complex industrial environments.

Area of Work: Anomaly Detection, Synthetic Data Generation

3. Objectives

- ⇒ **Develop Robust Anomaly Detection:** Implement and validate VAEs to accurately identify anomalies in industrial data, specifically overcoming the challenge of scarce real anomalous examples.
- ⇒ **Generate Synthetic Industrial Data:** Implement GANs to produce realistic and diverse synthetic datasets that mimic complex operational scenarios and rare failure events.
- ⇒ **Enhance AI Model Performance:** Leverage the generated synthetic data to train and improve the generalization and robustness of downstream ML models for critical industrial tasks.

4. Scope of Work

1. In-Scope

- **Applications of GenAI Models:** Implementation and experimentation with VAEs and GANs on industrial data.
- **Anomaly Detection:** Training VAEs on *normal* industrial data to learn its underlying distribution and use the model to predict anomalous data points.
- **Synthetic Data Generation:** Training GANs to synthesize high-fidelity, diverse industrial data including failure scenarios.
- **Model Evaluation:** Implementation of model evaluation techniques for anomaly detection and rigorous quantitative & qualitative evaluation of the realism and diversity of the generated synthetic data.
- **ML Model Enhancement via Synthetic Data:** Utilizing the generated synthetic data to train and improve the generalization, robustness, and performance of existing, discriminative ML models.

2. Out-of-Scope

- **Transformer Models:** Implementation and application of Transformer based models for interaction or data generation are explicitly outside the scope.
- **Advanced HMI Development:** Designing and constructing full-scale, highly interactive Human-Machine Interfaces (HMI) or complete control systems based on GenAI outputs is outside the scope.

5. Detailed Plan of Work

Sl. No.	Tasks/Subtasks to be done	Start Date - End Date	Planned Duration (weeks)	Specific Deliverable
Phase 1: Project Setup & Data Wrangling				
1	Project Initiation, Literature Review & Data Identification	Week 1 – Week 2	2	Proposal Approval, Literature Review, Dataset Finalization
2	Data Acquisition, Preprocessing & Splitting	Week 3 – Week 4	2	Clean & Prepare Dataset, Data Statistics Report
Phase 2: VAE Development for Anomaly Detection				
3	VAE Model Development, Training & Optimization	Week 5 – Week 6	2	Implement VAE Models, Implement Anomaly Scoring Function, Model Evaluation
4	VAE Performance Evaluation	Week 7 – Week 8	2	VAE Anomaly Detection Performance Report
Phase 3: GAN Model Development & Synthetic Data Generation				
5	GAN Model Development, Training & Optimization	Week 9 – Week 10	2	Implement GAN Models, Model Evaluation
6	GAN Model Comparison & Final Dataset Creation	Week 11 – Week 12	2	Model Comparison Report, Generate Synthetic Datasets
Phase 4: Dataset Augmentation & ML Model Generalization				
7	ML Model Enhancement via Synthetic Data	Week 13 – Week 14	2	ML Model Performance Comparison Report
Phase 5: Dissertation Finalization & Submission				
8	Overall Project Evaluation	Week 15 – Week 15	1	Comprehensive Evaluation Report
9	Dissertation Finalization & Submission Preparation	Week 16 – Week 16	1	Complete Dissertation Documentation and Submit (Final)

6. Literature Review

References

- [1] Radicioni, L.; Bono, F.M.; Cinquemani, S. “*Vibration-Based Anomaly Detection in Industrial Machines: A Comparison of Autoencoders and Latent Space*”. *Machines* 2025, 13, 139. <https://doi.org/10.3390/machines13020139>
- [2] A. A. Pol, V. Berger, C. Germain, G. Cerminara and M. Pierini, “*Anomaly Detection with Conditional Variational Autoencoders*”, 2019 18th IEEE International Conference On Machine Learning And Applications (ICMLA), Boca Raton, FL, USA, 2019, pp. 1651-1657, doi: 10.1109/ICMLA.2019.00270. <https://ieeexplore.ieee.org/document/8999265>
- [3] Keskes, Mohamed. (2025). “*Generative Adversarial Networks for Synthetic Data Generation in Deep Learning Applications*”, Preprints, 2025. 10.20944/preprints202505.0836.v1. https://www.researchgate.net/publication/391710194_Generative_Adversarial_Networks_for_Synthetic_Data_Generation_in_Deep_Learning_Applications
- [4] H. Lu, M. Du, K. Qian, X. He and K. Wang, “*GAN-Based Data Augmentation Strategy for Sensor Anomaly Detection in Industrial Robots*” in *IEEE Sensors Journal*, vol. 22, no. 18, pp. 17464-17474, 15 Sept.15, 2022, doi: 10.1109/JSEN.2021.3069452. <https://ieeexplore.ieee.org/abstract/document/9389530>

Supervisor's Rating of the Technical Quality of this Dissertation Outline

Rating (EXCELLENT / GOOD / FAIR/ POOR) : EXCELLENT



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